Sparse Eisenstein radix on secp256k1: exact recoding and paired panel

Question. Does the table-light radix $\pi = 127 + 128\tau$ reproduce the frozen secp256k1 scalar stream, and how does its prepared-table online time compare with the earlier 334−τ Horner path?

Scope and identity. The curve is SEC 2 secp256k1, $E: y^2 = x^3 + 7$ over $p = 2^{256} - 2^{32} - 977$, using the prime-order subgroup n = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141 generated by G. No immutable EC1 curve/subgroup ID for secp256k1 exists in the current curve catalog, so this result uses the standard exact curve and subgroup parameters rather than inventing a registry ID. The scalar fixture contains 12,288 distinct SHAKE256-derived scalars; its SHA-256 is a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3.

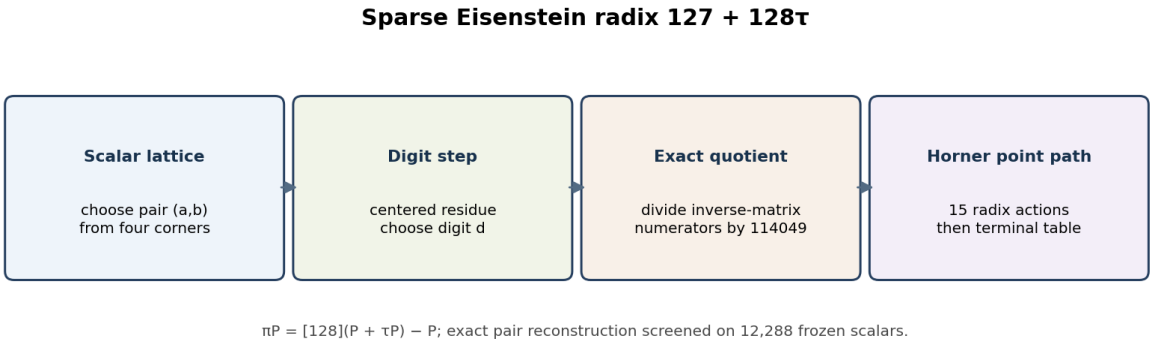


Figure 1: Sparse radix recoding and native evaluation path

Derived pair arithmetic. In the basis $1,\tau$ with $\tau^2 = 3\tau - 3$, the radix norm is $N(127+128\tau)=114049$; its quotient residue is $\tau \equiv -892 \pmod{114049}$. The pair multiplication matrix is $(a,b) \mapsto (127a-384b, 128a+511b)$. Since $127+128\tau = 128(1+\tau)-1$, its point action is $\pi P=[128](P+\tau P)-P$, implemented as one τ action, seven doublings, then subtraction of P. These identities are checked by sparse_pair_shortcut, sparse_norm, tau_residue, and sparse_pair_norm in SparseTauRadix.lean.

For a pair (A,B) after subtracting the selected digit, the inverse matrix gives quotient numerators $511A+384B$ and $-128A+127B$, each divided by 114049. The native selector enumerates nearby residue lifts and nearby integer candidates around the norm-minimizing center, then chooses the lexicographic minimum (norm, |b|, |a|, a, b). The independent screen exhausts all 57,025 signed residue indices and recodes the frozen 12,288-scalar fixture for 15 steps, checking exact integer divisibility and reconstruction at each step. Rerunning it reproduced sparse-tau-radix-screen.json byte-for-byte: maximum selected digit norm 37,804, maximum observed terminal norm 7, and 25 distinct terminal pairs. This is finite fixture evidence; the theorem linking the pair action to an actual curve-wide group law is conditional in Lean and remains open.

Native implementation and measurement. I inspected the exact provenance archive implementation. Its residue folding, digit sign handling, inverse-matrix quotient, and divisibility assertions match the independent integer formulas; its point multiplier matches the Lean pair identity. The pinned Lean check proves the pair identities and exact trace reconstruction, then states the point-Horner result under explicit map_add and map_radix hypotheses. Those hypotheses are not a formal refinement proof of the Rust field and curve operations.

The retrospective serial Tier 1 panel used one RunPod instance (thyvmdqmzsz5ik, CPU 40), 16 alternating repetitions, 96 arms, and 48 paired records. Each timed arm evaluated the same 12,288 distinct scalars after per-process table preparation; fixture loading and independent output checks were outside Rust Instant. The manifest, frozen rules, launch receipt, auditor, raw panel archive, and native provenance retain the exact run. Rerunning the auditor reproduced tier1-sparse-tau-cpu40-audit.json byte-for-byte: all 96 arms and 48 pairs were valid, artifacts were stable, all A/A controls passed, and the paired interval excluded 1.

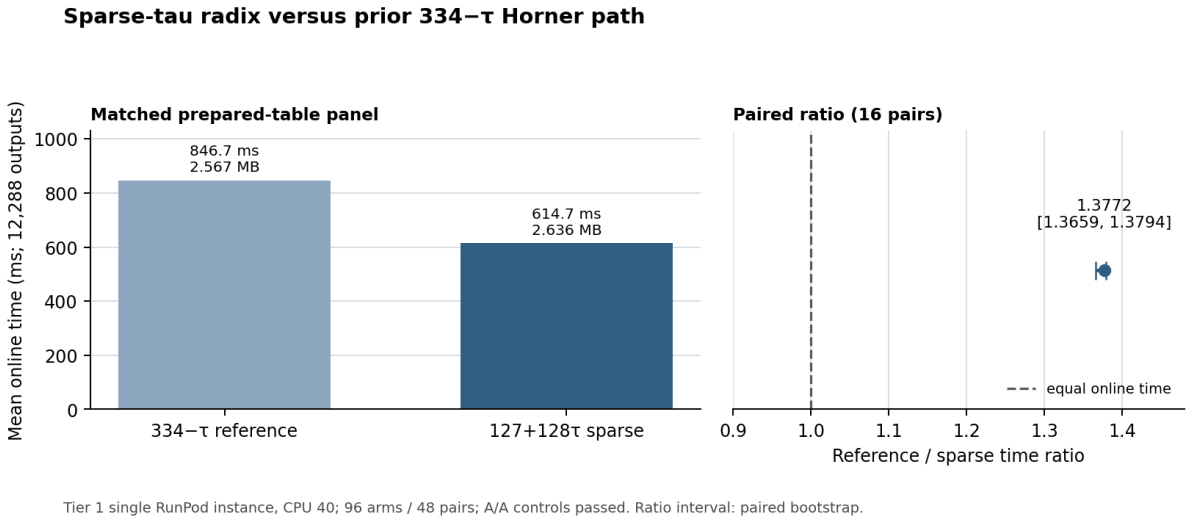


Figure 2: Prepared-table means and qualified paired ratio

Prepared-table path	Mean online time	Retained table
334−τ reference	846.675290 ms	2,566,612 bytes
127+128τ sparse	614.672321 ms	2,636,452 bytes

The qualified reference/sparse paired median is `1.3771539`, with the predeclared 95% by-repeat bootstrap interval `[1.3659075, 1.3794007]` across 16 valid comparison pairs. This supports an online-time advantage for this exact prepared-table workload on this observed cloud instance. The sparse table retains 69,840 more bytes. It does not establish cold single-scalar latency or performance on other hardware; those require separately timed panels.

Negative findings and limits. The exact digit screen proves neither optimality of the bounded selector over every lattice point nor a formal implementation refinement. The finite replay covers the committed fixture, not every subgroup scalar. The `point_replay_status` field in the recoding screen remains `open_native_group_check`; do not promote the screen to a curve-wide theorem. The native panel's output verification is independent finite evidence, not a substitute for the missing curve-wide group-law and arithmetic bridge. The shared sparse-radix correctness problem record tracks that proof gap. No new curve, endomorphism, isogeny, or candidate route was found. The study tree has no canonical curve/isogeny graph source or rendering to update; the main catalog was checked and contains no secp256k1 record, so no graph edge or curve entry is changed.

The editable figure source reads the committed audit JSONs; its SVG/PNG outputs and this report's PDF accompany the report. The report presents a measured single-instance result while keeping the exact unresolved proof obligations visible.